

Routing Essentials

Tecnologie e Servizi di Rete
Computer Network Technologies and Services

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*part of the slides adapted from material by Jim Kurose



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Outline

- Introduction and Taxonomy
 - Centralized routing
 - Distributed routing
 - Distance Vector algorithms
 - Link State algorithms

- Internet Routing Architecture
 - Intra-AS Routing protocols
 - Inter-AS Routing

Introduction and Taxonomy

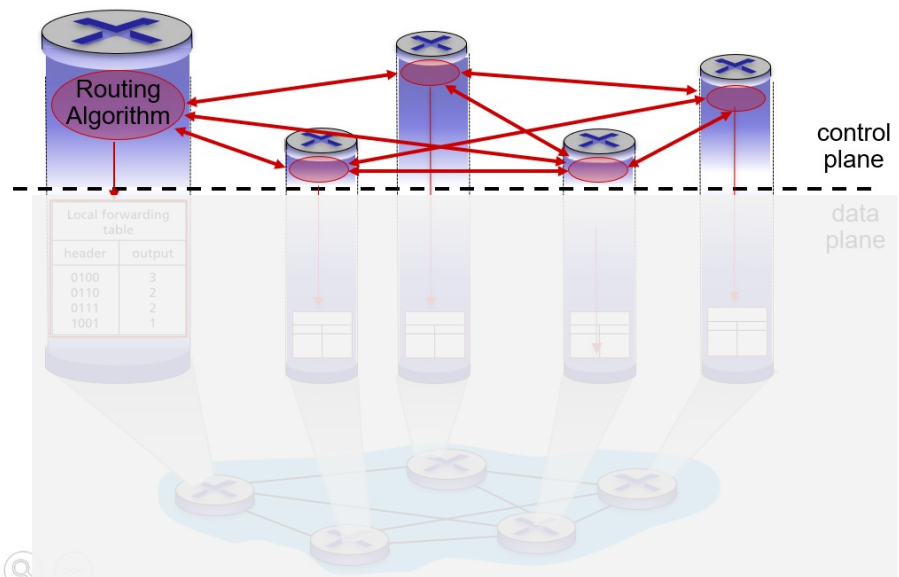


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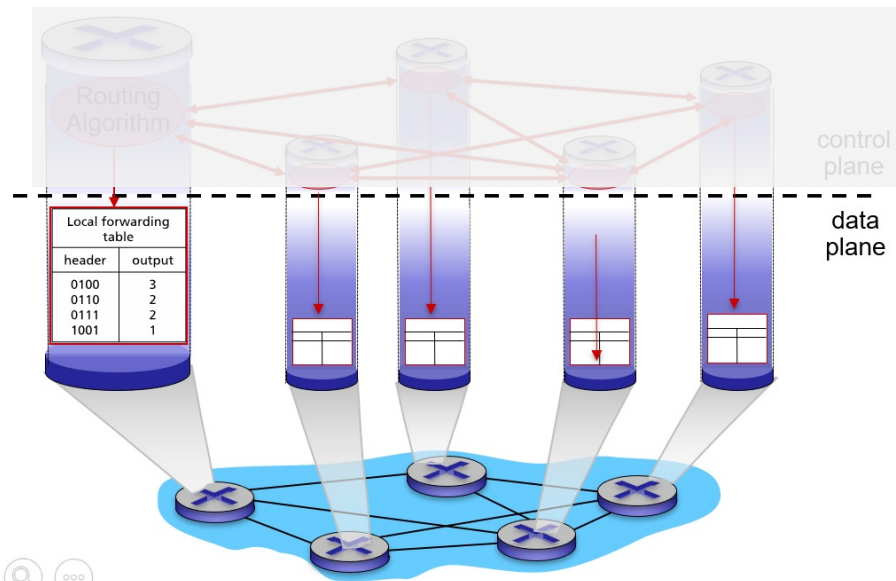
Routing

- Determination of (optimal) path from source to destination for packet transmission in a computer network
- Based on routing algorithms and protocols
- It is a *control plane* operation



Forwarding

- Transmission of packets from one network device (router) to the next hop in their route
 - no real-time decision-making, just routing table look-up
 - next-hop choice predetermined by a routing algorithm and written in the routing table
- It is a *data plane* operation



Forwarding Procedures

- Forwarding by Network Address (most common)
- Label Swapping
 - e.g., MPLS (see later)
- Source Routing (rarely used)

Forwarding Phases

- Next-hop/output port selection (using routing tables)
- Switching: transfer to output port
- Transmission

Routing Algorithm Classification

- Non-adaptive algorithms (static)
- Adaptive algorithms (dynamic)

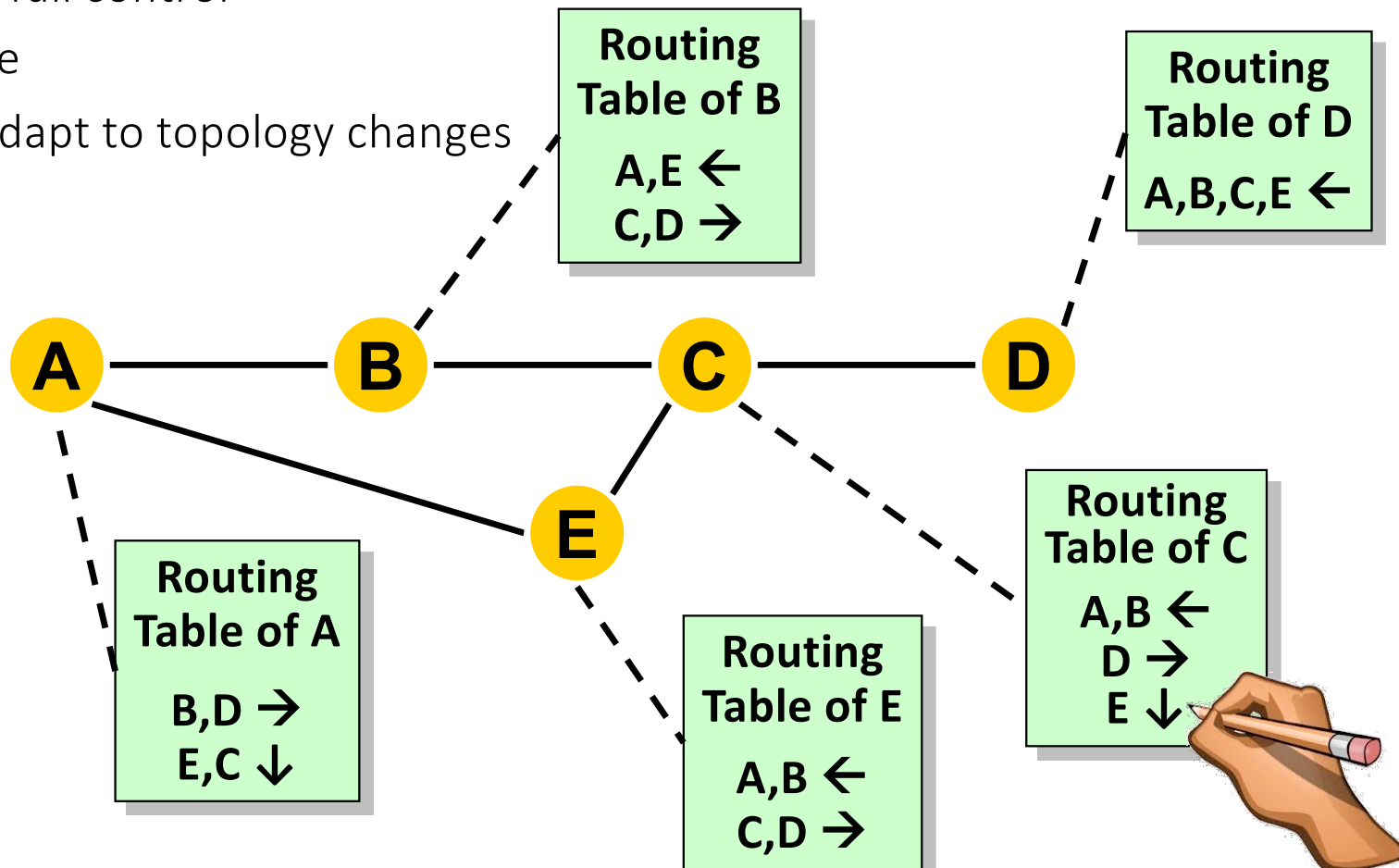
Non-adaptive Algorithms

- Fixed Directory routing
 - Static routing
 - Manual configuration
- (Selective) flooding and similar
 - Random
 - Randomly select an output port
 - Flooding
 - Forward on any output port (except the one on which the PDU was received)
 - Deflection or «hot potato»
 - Used on regular topologies
 - If the correct port is available, route on it. If busy, select an available (non optimal) port



Fixed Directory Routing

- Admin has full control
- Error prone
- does not adapt to topology changes



Adaptive Algorithms

- Centralized Routing
- Isolated Routing
- Distributed Routing
 - Distance Vector
 - Link State



Centralized Routing

- One node (Routing Control Center, RCC) collects info from all other nodes
- Computes the path
- Distributes to all nodes

Centralized Routing: PROs

- Routing Control Center (RCC)
 - Calculates and distributes routing tables
 - Needs information from all nodes
 - Optimizes performance
 - Simplifies troubleshooting



Centralized Routing: CONs

- RCC is single point of failure
- RCC is bottleneck
 - Significant network load in proximity of RCC
- Not suitable for highly dynamic networks



Isolated Routing

- Each node decides independently
- No exchange of information
- E.g., Backward Learning
 - IEEE 802.1D switches



Distributed Routing

- Routers co-operate in exchanging connectivity information
- Each router decides independently, but **coherently**
- Combines advantages of isolated and centralized routing
- Two big families:
 - Distributed Routing with Partial Information
 - Distance Vector algorithms
 - Distributed Routing with Global Information
 - Link State algorithms



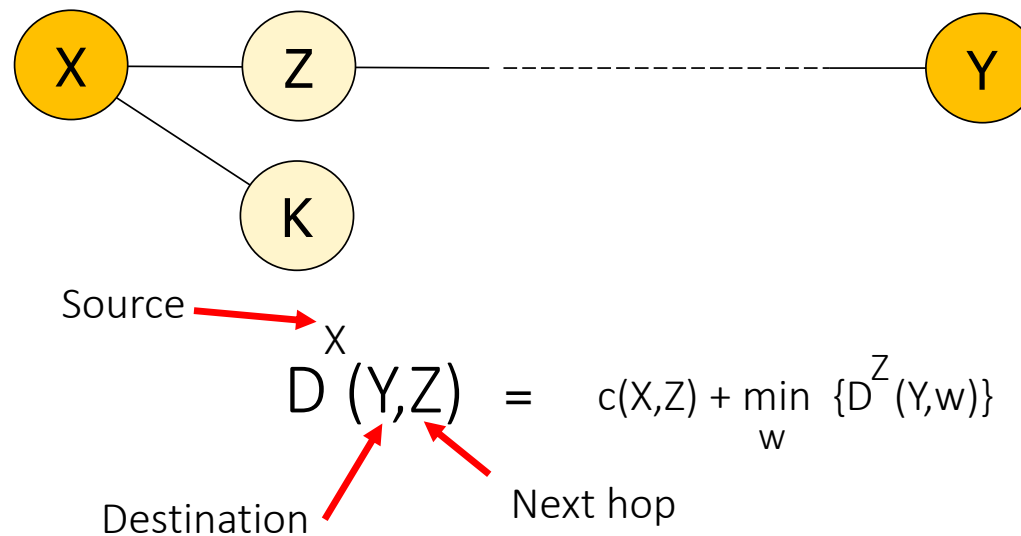
Distance Vector Algorithm

- Every node periodically exchanges with adjacent nodes a **Distance Vector**, containing
 - Reachable destinations
 - Current distance to each destination
 - E.g., number of hops on the path
- Each node maintains two data structures:
 - **Distance Table (DT)**: lists costs to any known destination **through each neighbor (next hop)**
 - **Routing Table (RT)**: lists costs and next hops of **selected** routes to each destination
- A Distance Vector contains the same information as the RT, minus the next hop
- Nodes compare the received vector with the current DT and RT and modify them
 - Adding new destinations
 - Modifying routing if new paths are shorter
 - Modifying costs if needed
- Bellman–Ford algorithm

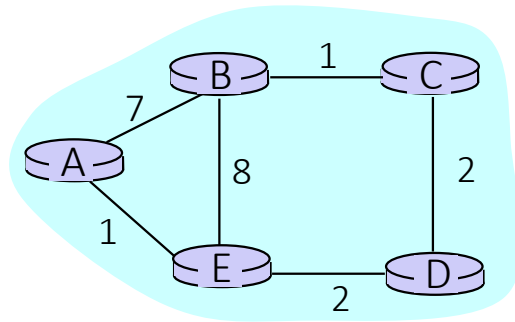


Distance Vector: implementation

- Structure of Distance Table in every node:
 - One row for each possible destination
 - One column for each adjacent node
- Example: Element of node X's DT, for destination Y through adjacent node Z
 - Cost of reaching Y from X, using Z as next hop



Distance Table: example



$$\begin{aligned} D^E(C,D) &= c(E,D) + \min_w \{D^D(C,w)\} \\ &= 2+2 = 4 \end{aligned}$$

$$\begin{aligned} D^E(A,D) &= c(E,D) + \min_w \{D^D(A,w)\} \\ &= 2+3 = 5 \quad \text{loop!} \end{aligned}$$

$$\begin{aligned} D^E(A,B) &= c(E,B) + \min_w \{D^B(A,w)\} \\ &= 8+6 = 14 \quad \text{loop!} \end{aligned}$$

Cost to .. Via ...

$D^E()$	A	B	D
A	1	14	5
B	7	8	5
C	6	9	4
D	4	11	2

destination



From DT to RT

Cost to .. Via ...

$D^E()$	A	B	D
A	1	14	5
B	7	8	5
C	6	9	4
D	4	11	2

Destination Table



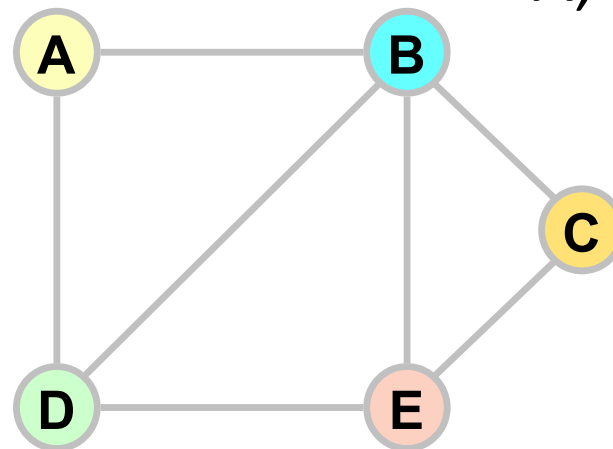
	Next Hop, Cost
A	A,1
B	D,5
C	D,4
D	D,2

Routing Table



Sample Scenario

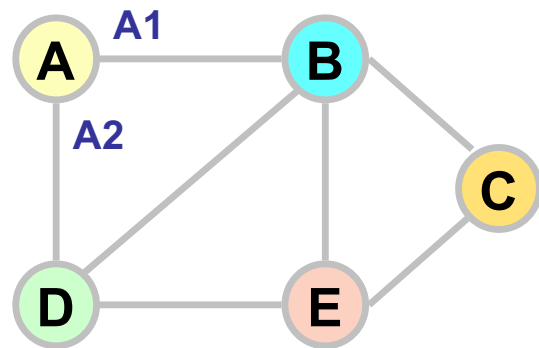
Routing information
stored by A



<u>Loc (A)</u>	<u>DV (B)</u>	<u>DV (D)</u>
A, 0	A, 1	A, 1
	B, 0	B, 1
	C, 1	C, 2
	D, 1	D, 0
	E, 1	E, 1



Distance Vector Merging and Generation



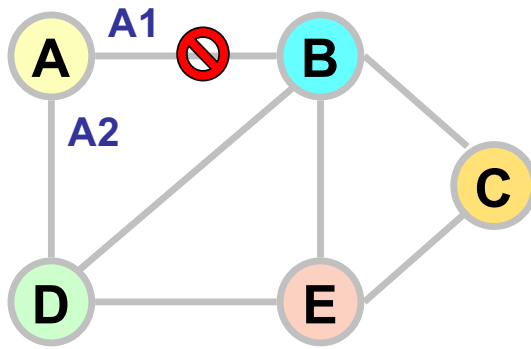
Received from line A1

Received from line A2

<u>Loc (A)</u>	<u>DV (B)</u>	<u>DV (D)</u>	<u>ROUT. TABLE (A)</u>	<u>DV (A)</u>
<u>A, 0</u>	A, 1	A, 1	A, local, 0	A, 0
	<u>B, 0</u>	B, 1	B, A1, 1	B, 1
	<u>C, 1</u>	C, 2	C, A1, 2	C, 2
	D, 1	<u>D, 0</u>	D, A2, 1	D, 1
	E, 1	<u>E, 1</u>	E, A2, 2	E, 2



Topology Change



<u>Loc (A)</u>	<u>DV (B)</u>	<u>DV (D)</u>	<u>ROUT. TABLE (A)</u>	<u>DV (A)</u>
<u>A, 0</u>	A, 1	A, 1	A, local, 0	A, 0
	B, 0	B, 1	B, A2, 2	B, 2
	C, 1	C, 2	C, A2, 3	C, 3
	D, 1	D, 0	D, A2, 1	D, 1
	E, 1	E, 1	E, A2, 2	E, 2



Example: Cold Start

<u>RT</u> (A)	<u>RT</u> (B)	<u>RT</u> (C)	<u>RT</u> (D)	<u>RT</u> (E)
A, loc, 0	B, loc, 0	C, loc, 0	D, loc, 0	E, loc, 0

A sends its DV

<u>RT</u> (A)	<u>RT</u> (B)	<u>RT</u> (C)	<u>RT</u> (D)	<u>RT</u> (E)
A, loc, 0	A, B1, 1	C, loc, 0	A, D1, 1	E, loc, 0
	B, loc, 0		D, loc, 0	

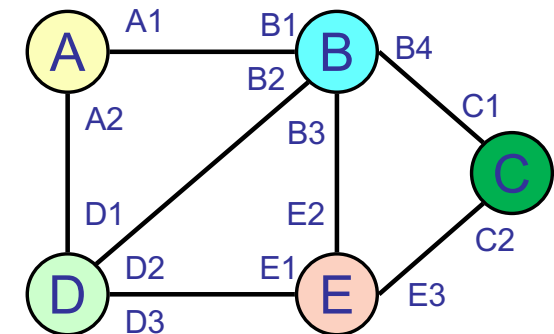
B and D send their DVs

<u>RT</u> (A)	<u>RT</u> (B)	<u>RT</u> (C)	<u>RT</u> (D)	<u>RT</u> (E)
A, loc, 0	A, B1, 1	A, C1, 2	A, D1, 1	A, E2, 2
B, A1, 1	B, loc, 0	B, C1, 1	B, D2, 1	B, E2, 1
D, A2, 1	D, B2, 1	C, loc, 0	D, loc, 0	D, E1, 1
				E, loc, 0

All send their DVs

. . .

<u>RT</u> (A)	<u>RT</u> (B)	<u>RT</u> (C)	<u>RT</u> (D)	<u>RT</u> (E)
A, loc, 0	A, B1, 1	A, C1, 2	A, D1, 1	A, E2, 2
B, A1, 1	B, loc, 0	B, C1, 1	B, D2, 1	B, E2, 1
C, A1, 2	C, B4, 1	C, loc, 0	C, D2, 2	C, E3, 1
D, A2, 1	D, B2, 1	D, C2, 2	D, loc, 0	D, E1, 1
E, A2, 2	E, B3, 1	E, C2, 1	E, D3, 1	E, loc, 0



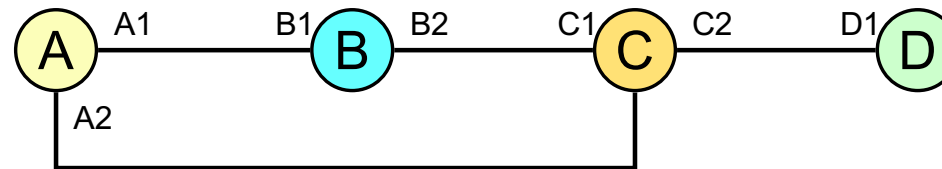
Issues

- Black Hole
 - DoS attack by malicious node advertising impossibly short, non-existent routes to attract traffic, which is then discarded
- Count to infinity
 - Misalignment of Routing Tables due to slow convergence time of the Distance Vector algorithm and inconsistent information at different nodes
- Bouncing Effect
 - (Temporary) Loop caused by inconsistent Routing Tables following a link failure

Common Reason: lack of information on the global topology

Path Vector

- Nodes exchange dynamically-updated path information
- Easy to detect loops in the vector
- Increased overhead



A:

Loc (A)
A, 0

PV (B)
A, 1, [B]
B, 0, [-]
C, 1, [B]
D, 2, [B,C]

PV (C)
A, 1 [C]
B, 1, [B]
C, 0, [-]
D, 1, [D]



RT (A)
A, loc, 0
B, A1, 1
C, A2, 1
D, A2, 2



PV (A)
A, 0, [-]
B, 1, [A]
C, 1, [A]
D, 2, [A,C]

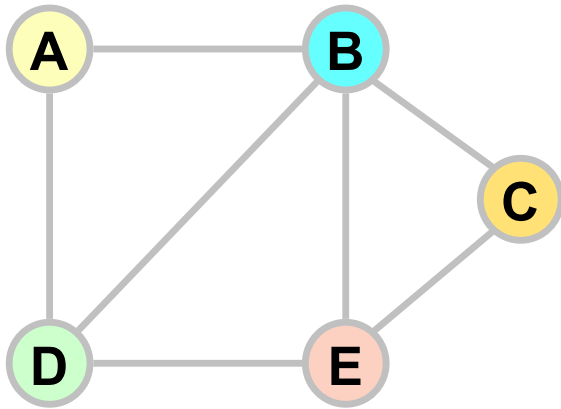


Link State algorithm

- Every node sends cost info (status) of each link in broadcast to all other nodes
 - Implemented using multicast among routers
- Every nodes builds the network topology, knowing all link costs
- Every node computes minimum cost path (paths) toward every other node
 - info stored in the routing table
- Dijkstra algorithm is used to compute the optimal path
 - Iterative algorithm: after k iterations the minimum cost path toward k destinations are obtained
 - Only works for positive costs



Link State Database



Stored by each Node

<u>LS (A)</u>	<u>LS (B)</u>	<u>LS (C)</u>	<u>LS (D)</u>	<u>LS (E)</u>
B, 1	A, 1	B, 1	A, 1	B, 1
D, 1	C, 1	E, 1	B, 1	C, 1
	D, 1		E, 1	D, 1
	E, 1			



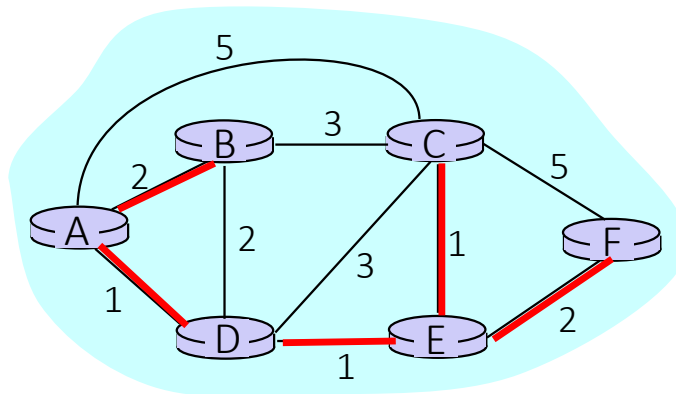
Dijkstra Algorithm

- Low complexity: $L * \log(N)$
 - L: number of links
 - N: number of nodes
- Shortest Path First
 - The next node “nearest” to the root is identified
 - Its next hop is inserted into the routing table



Dijkstra: example

Step	start N	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
→ 0	A	2,A	5,A	1,A	infinity	infinity
→ 1	AD	2,A	4,D		2,D	infinity
→ 2	ADE	2,A	3,E			4,E
→ 3	ADEB		3,E			4,E
→ 4	ADEBC					4,E
5	ADEBCF					



	Next hop, cost
B	B,2
C	D,3
D	D,1
E	D,2
F	D,4



PROs

- Rapid Convergence
 - LSs spread quickly
 - No intermediate processing
- Limited Routing Traffic and Storage
 - Link states are small
 - Fast and efficient neighbor greeting
- It rarely generates loops
- Simple to understand and troubleshoot
 - All nodes have identical databases



CONS

- High implementation complexity
 - Selective flooding
 - First implementation took several years
- Protocols with complex configuration



Link State Generation

- In principle: when there is a topology change
- Actual protocols: LS are generated periodically
 - Increased reliability

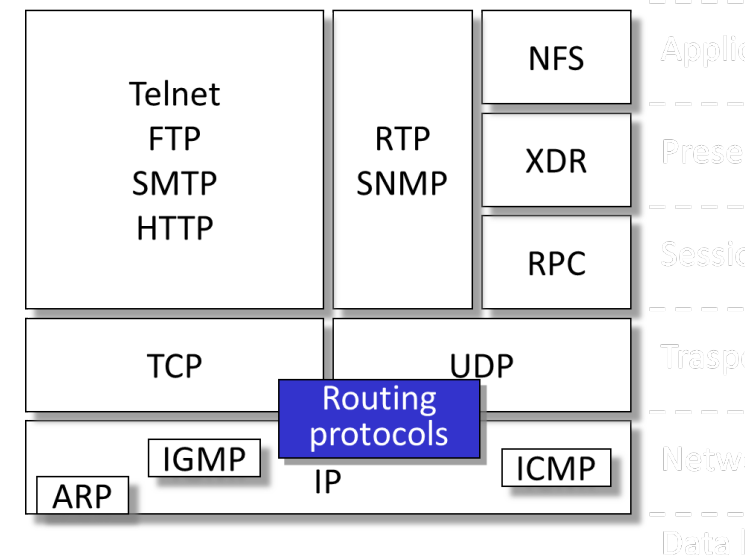


Internet Routing Architecture



Routing Protocols

- Protocols that allow routers to exchange information on the network and determine the best route to each destination
 - Uses routing algorithms
- Define metric(s) and their encoding in packets
- Specific timing
- Problem: what is the operational domain of the protocol?
 - A subnet?
 - An operator's network?
 - A country?
 - The Internet at large?



Making routing scalable

scale: with billions of destinations:

- can't store all destinations in routing tables!
- routing table exchange would swamp links!
- our routing description thus far:
 - idealized
 - all routers identical
 - network “flat”
 - ... not true in practice

administrative autonomy

- internet = network of networks
- each network admin may want to control routing in its own network



Autonomous System

- A set of subnets grouped based on
 - Topology
 - Organizational criteria
- E.g., the subnets of a large internet service provider
- Administration
 - Autonomous internal routing choices
 - Negotiated external routing choices
- Scalability
 - Not all information propagated everywhere
- Identification: AS number
 - Two bytes
 - Assigned by IANA



Internet approach to scalable routing

Aggregate routers into “autonomous systems” (AS) (a.k.a. “domains”)

- intra-AS routing
 - routing among hosts, routers in same AS (“network”)
 - all routers in AS must run same intra-domain protocol
- routers in different AS can run different intra-domain routing protocol
- gateway router: at “edge” of its own AS, has link(s) to router(s) in other AS'es
- inter-AS routing
 - routing among AS'es
 - gateways perform inter-domain routing (as well as intra-domain routing)



Internet Routing Architecture

Intra-AS Routing

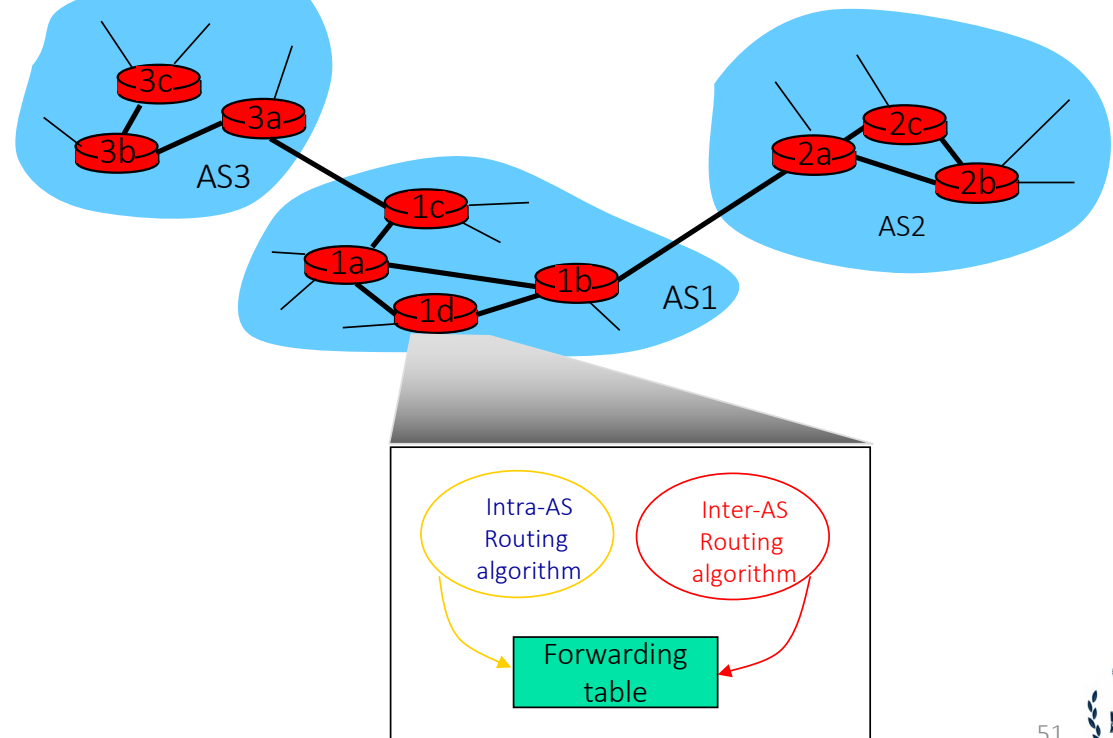


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Interconnected ASes

- forwarding table configured by both intra- and inter-AS routing algorithm
 - intra-AS routing determine entries for destinations within AS
 - inter-AS & intra-AS determine entries for external destinations



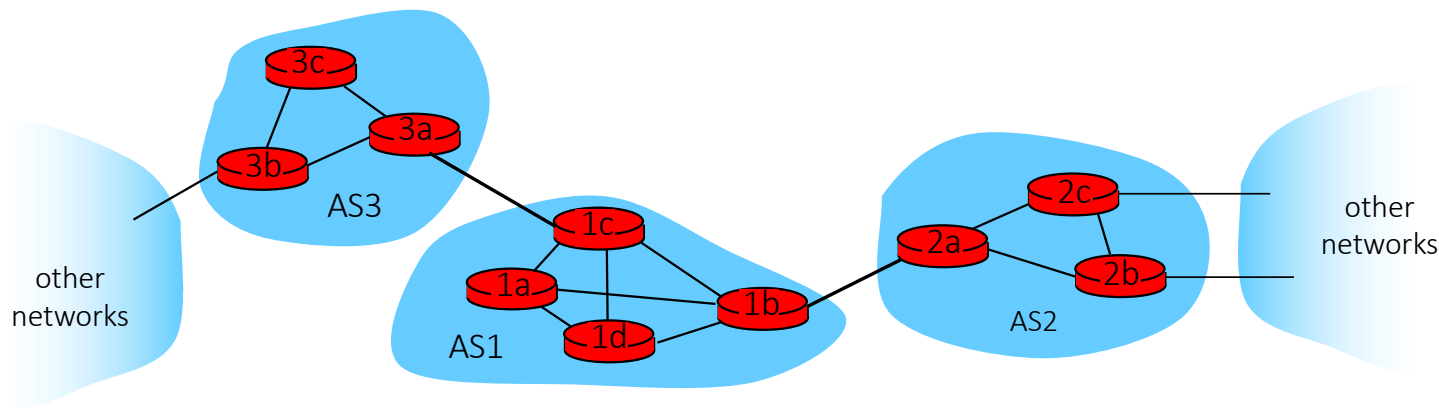
Inter-AS tasks

- suppose router in AS1 receives datagram destined outside of AS1:
 - router should forward packet to gateway router, but which one?

AS1 must:

1. learn which destinations are reachable through AS2, which through AS3
2. propagate this reachability info to all routers in AS1

job of inter-AS routing!



Intra-AS Routing

- also known as *interior gateway protocols (IGP)*
- most common intra-AS routing protocols:
 - RIP: Routing Information Protocol (obsolete)
 - OSPF: Open Shortest Path First (IS-IS protocol essentially same as OSPF)
 - IGRP: Interior Gateway Routing Protocol (Cisco proprietary for decades, until 2016)



OSPF (Open Shortest Path First)

- “open”: publicly available
- uses link-state algorithm
 - link state packet dissemination
 - topology map at each node
 - route computation using Dijkstra’s algorithm
- router floods OSPF link-state advertisements to all other routers in **entire** AS
 - carried in OSPF messages directly over IP (rather than TCP or UDP)
 - link state: for each attached link
- **IS-IS routing** protocol: nearly identical to OSPF

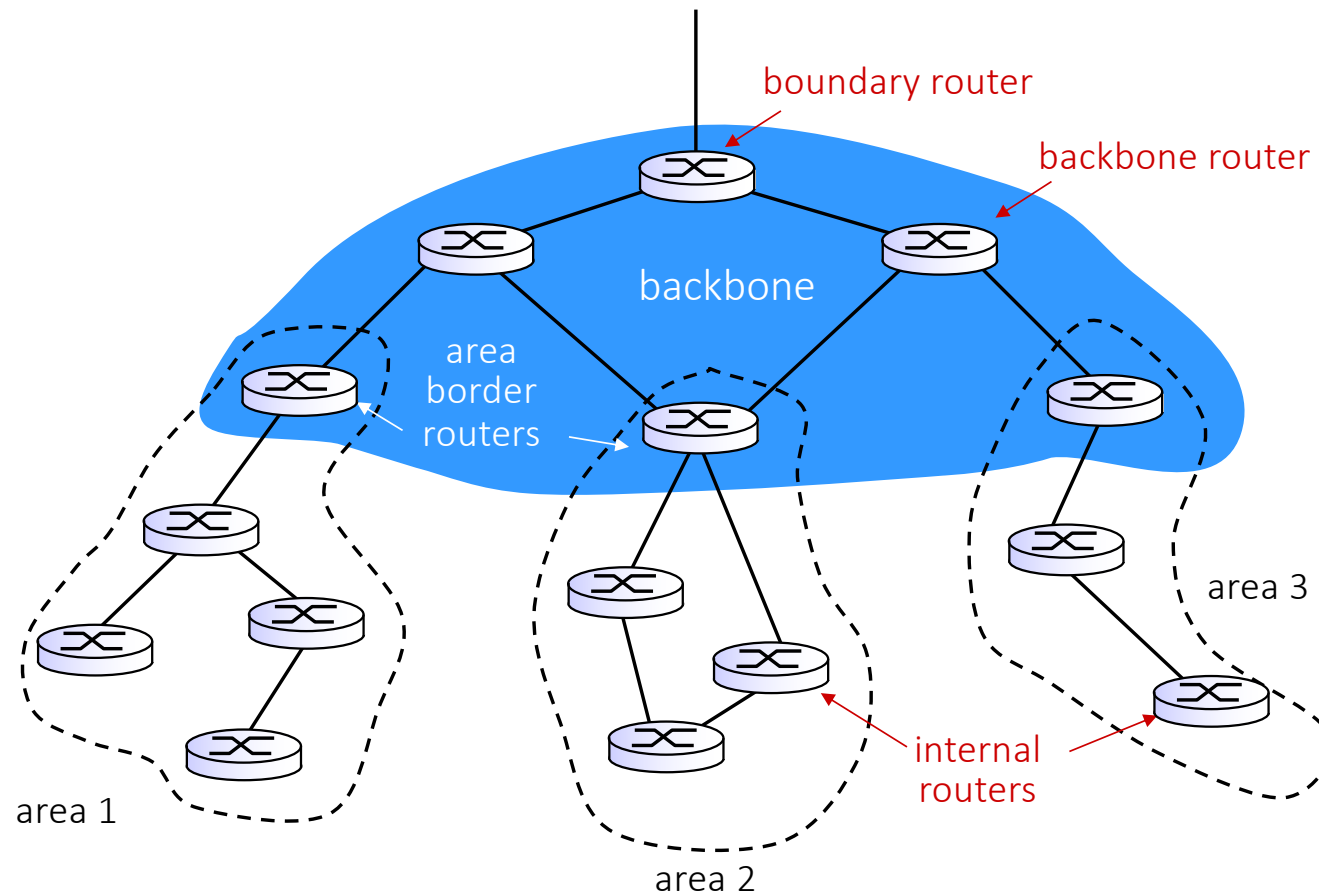


OSPF “advanced” features

- **security**: all OSPF messages authenticated (to prevent malicious intrusion)
- **multiple** same-cost **paths** allowed (only one path in RIP)
- for each link, multiple cost metrics for different **TOS** (e.g., satellite link cost set low for best effort ToS; high for real-time ToS)
- integrated uni- and **multi-cast** support:
 - Multicast OSPF (MOSPF) uses same topology database as OSPF
- **hierarchical** OSPF in large domains

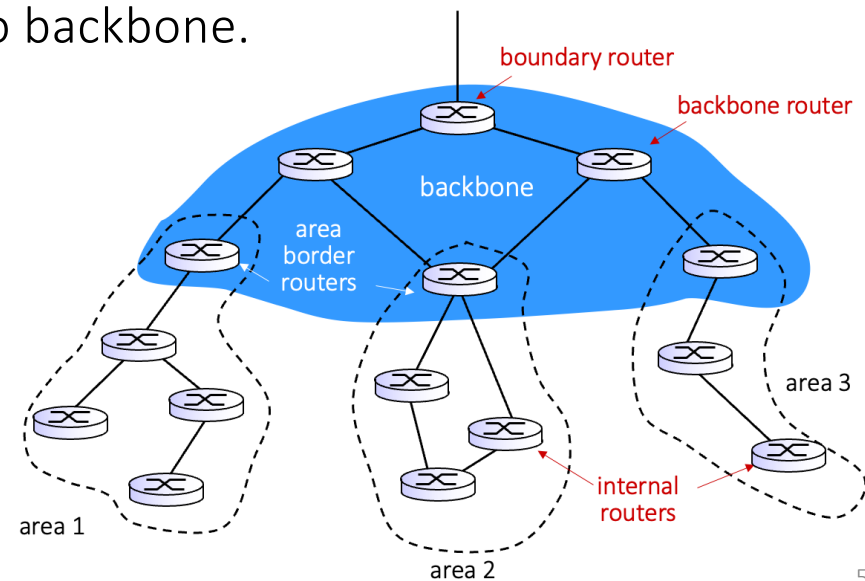


Hierarchical OSPF



Hierarchical OSPF

- **two-level hierarchy:** local area, backbone.
 - link-state advertisements only in area
 - each nodes has detailed area topology; only know direction (shortest path) to nets in other areas.
- **area border routers:** “summarize” distances to nets in own area, advertise to other Area Border routers.
- **backbone routers:** run OSPF routing limited to backbone.
- **boundary routers:** connect to other AS'es



Internet Routing Architecture

Inter-AS Routing



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Why different Intra-, Inter-AS routing ?

policy:

- inter-AS: admin wants control over how its traffic routed, who routes through its net.
- intra-AS: single admin, so no policy decisions needed

scale:

- hierarchical routing saves table size, reduced update traffic

performance:

- intra-AS: can focus on performance
- inter-AS: policy may dominate over performance

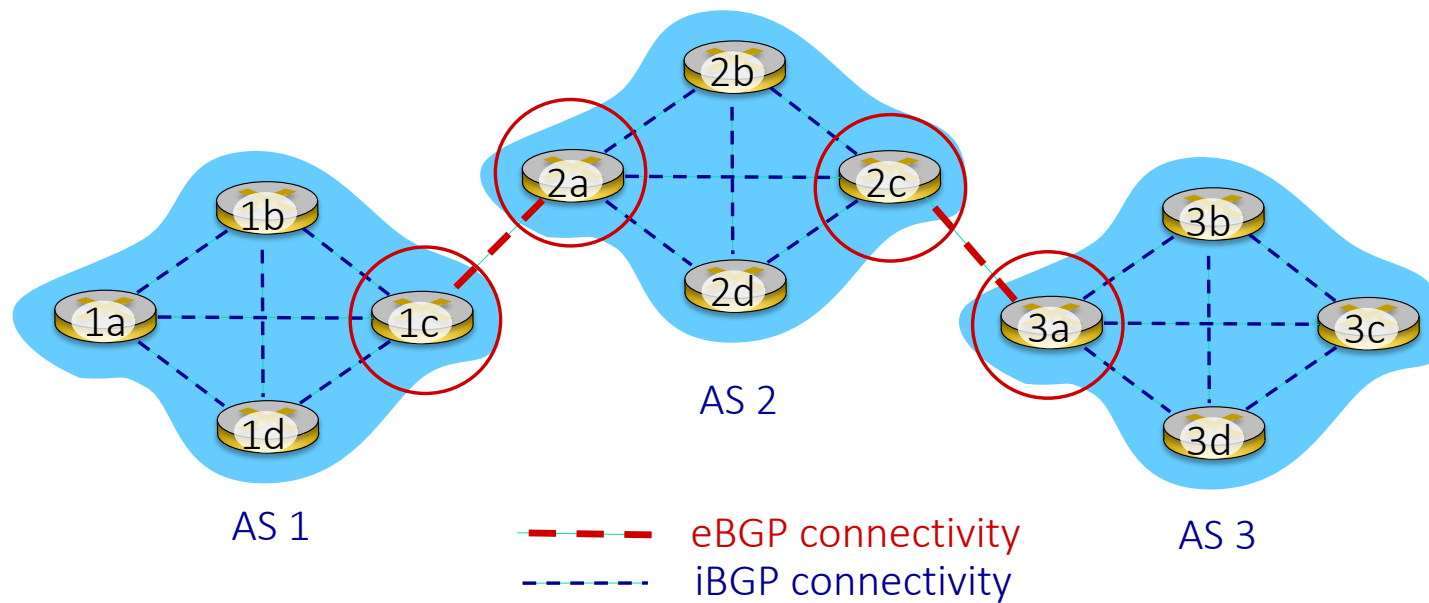


Internet inter-AS routing: BGP

- **BGP (Border Gateway Protocol):** the de facto inter-domain routing protocol
 - “glue that holds the Internet together”
- BGP provides each AS a means to:
 - obtain subnet reachability information from neighboring ASes (eBGP)
 - propagate reachability information to all AS-internal routers (iBGP)
 - determine “good” routes to other networks based on reachability information and policy
- Not necessarily shorter path
 - Choice based on policies
 - Reflect agreements among ASs
- allows subnet to advertise its existence to rest of Internet: “I am here”
- destinations can be aggregated
 - 195.1.2.0/24 and 195.1.3.0/24 can be announced as 195.1.2.0/23



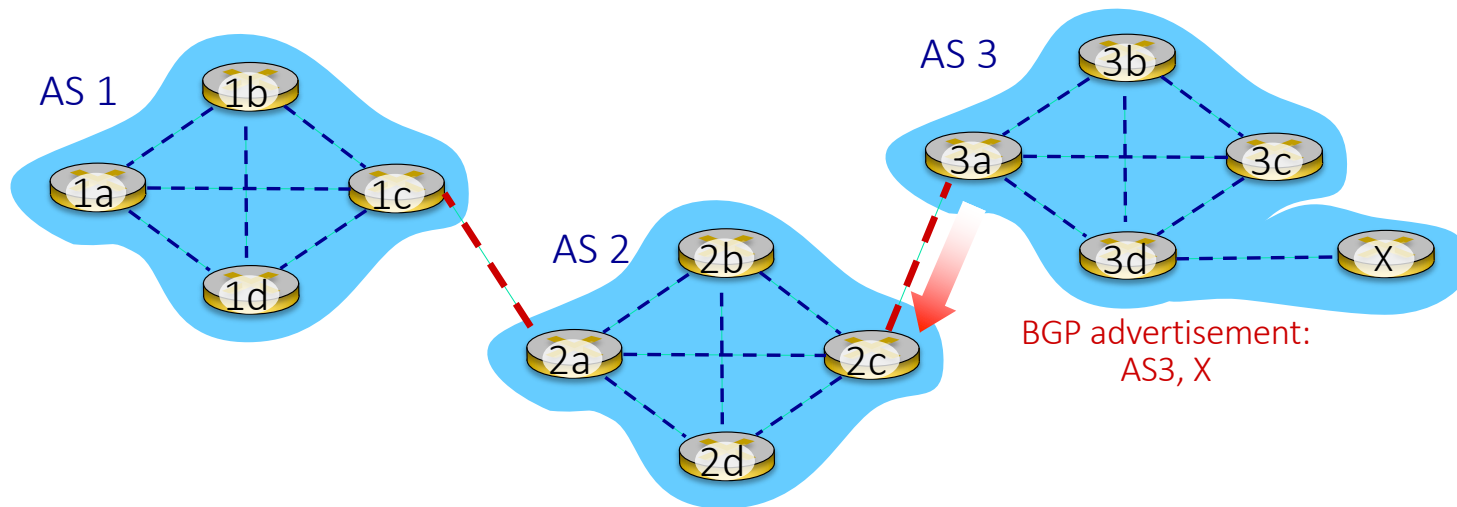
eBGP, iBGP connections



 gateway routers run both eBGP and iBGP protocols

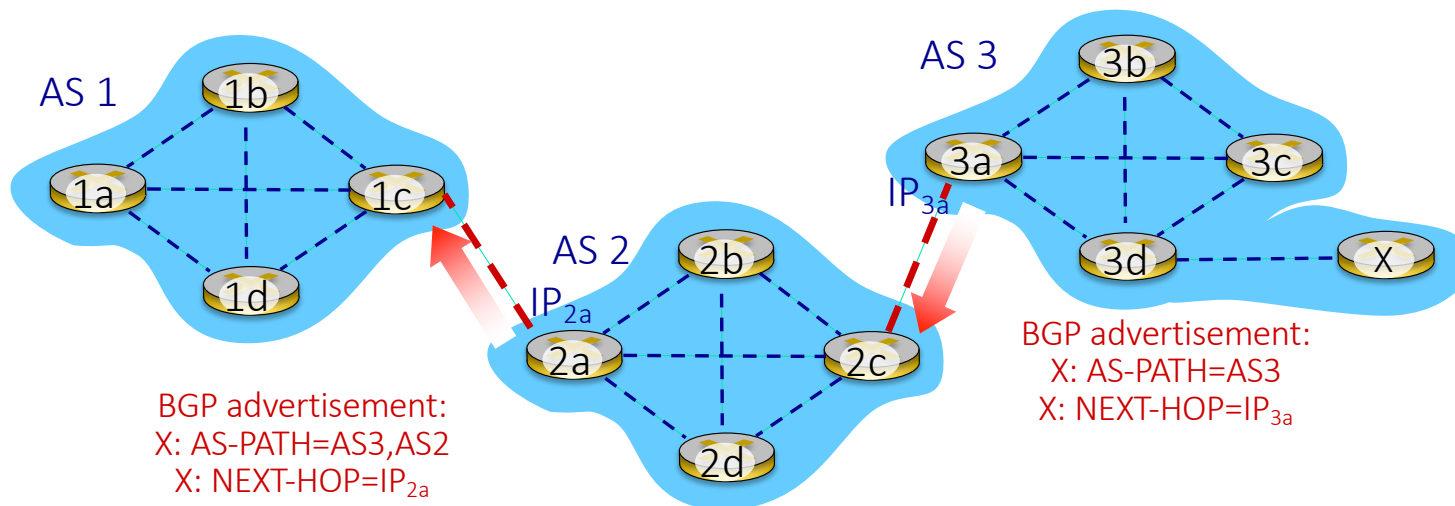
BGP basics

- **BGP session**: two BGP routers (“peers”) exchange BGP messages over semi-permanent TCP connection:
 - advertising **paths** to different destination network prefixes (BGP is a “path vector” protocol)
- when AS3 gateway router 3a advertises path **AS3,X** to AS2 gateway router 2c:
 - AS3 **promises** to AS2 it will forward datagrams towards X



Path attributes and BGP routes

- advertised prefix includes BGP attributes
 - prefix + attributes = “route”
- two important attributes:
 - **AS-PATH**: list of ASes through which prefix advertisement has passed
 - **NEXT-HOP**: indicates specific internal-AS router to next-hop AS



Path attributes and BGP routes

- advertised prefix includes BGP attributes
 - prefix + attributes = “route”
- two important attributes:
 - **AS-PATH**: list of ASes through which prefix advertisement has passed
 - **NEXT-HOP**: indicates specific internal-AS router to next-hop AS
- **Policy-based routing**:
 - gateway receiving route advertisement uses **import policy** to accept/decline path (e.g., never route through AS Y).
 - AS policy also determines whether to **advertise** path to other neighboring ASes



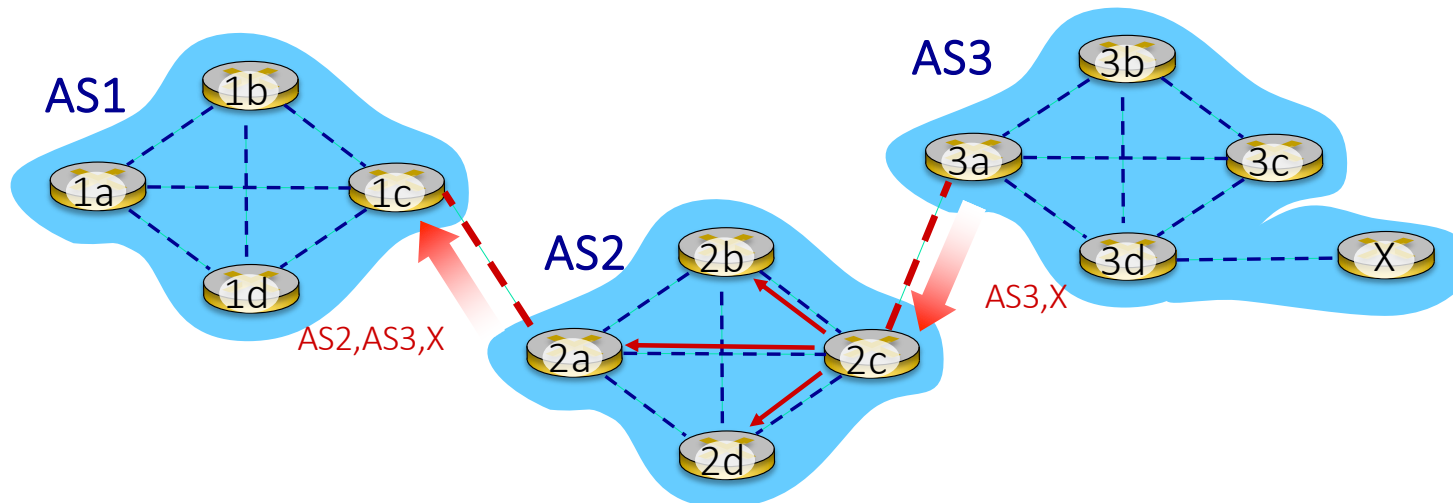
BGP messages

- BGP messages exchanged between peers over TCP connection
- BGP messages:
 - **OPEN**: opens TCP connection to remote BGP peer and authenticates sending BGP peer
 - **UPDATE**: advertises new path (or withdraws old)
 - **KEEPALIVE**: keeps connection alive in absence of UPDATES; also ACKs OPEN request
 - **NOTIFICATION**: reports errors in previous msg; also used to close connection



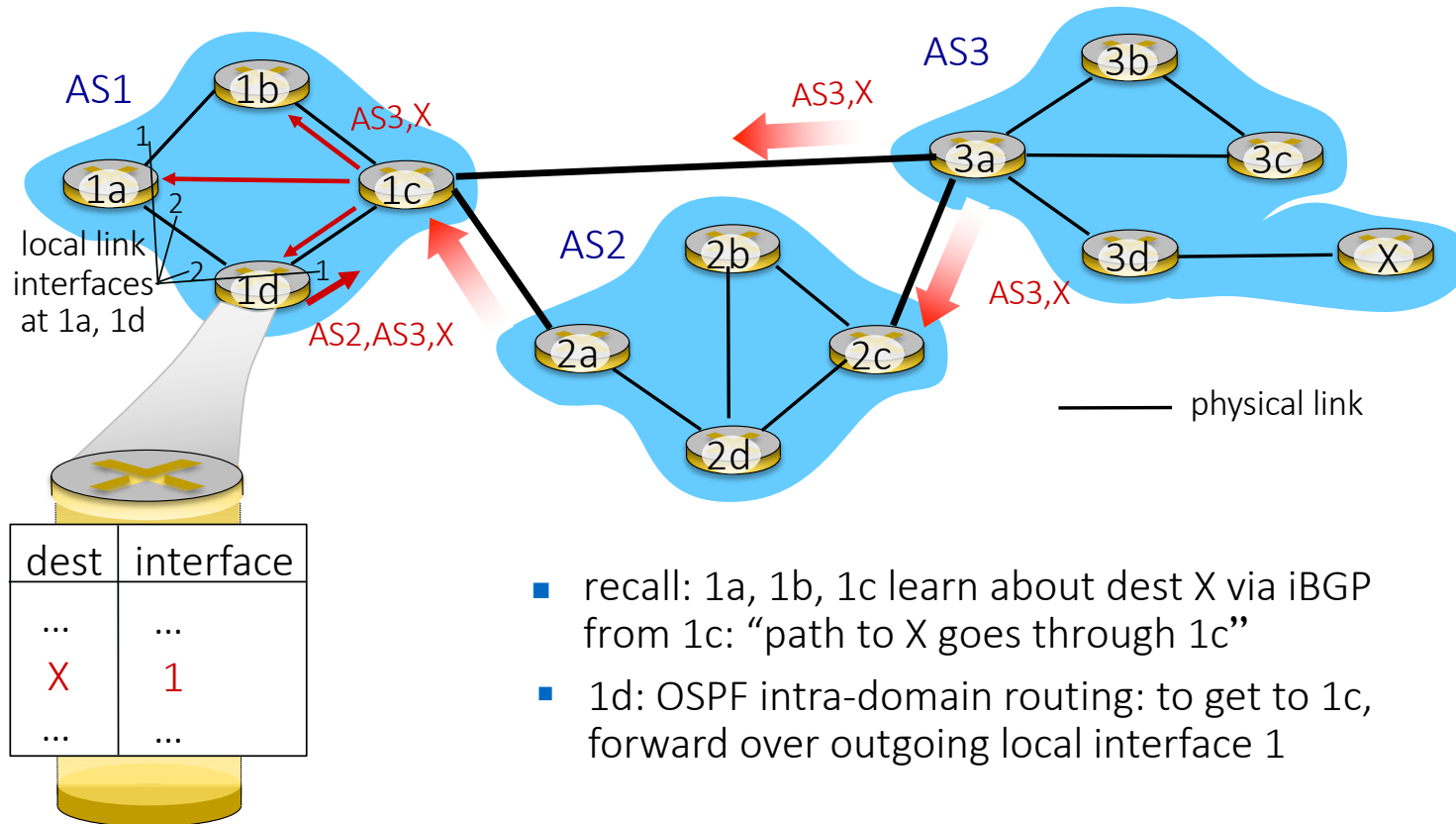
BGP path advertisement

- AS2 router 2c receives path advertisement **AS3,X** (via eBGP) from AS3 router 3a
- Based on AS2 policy, AS2 router 2c accepts path AS3,X, propagates (via iBGP) to all AS2 routers
- Based on AS2 policy, AS2 router 2a advertises (via eBGP) path **AS2, AS3, X** to AS1 router 1c



BGP, OSPF, forwarding table entries

Q: how does router set forwarding table entry to distant prefix?

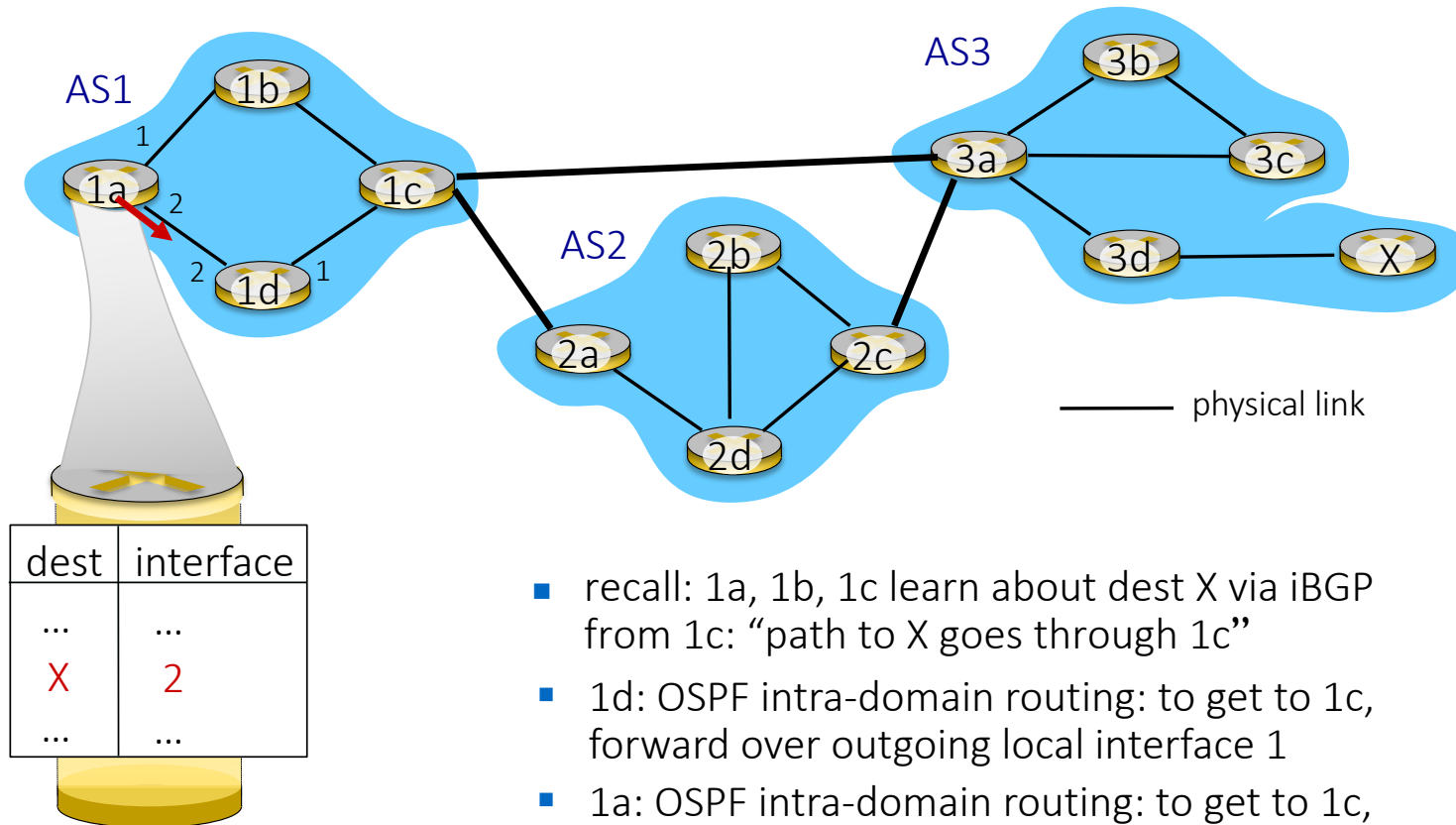


- recall: 1a, 1b, 1c learn about dest X via iBGP from 1c: “path to X goes through 1c”
- 1d: OSPF intra-domain routing: to get to 1c, forward over outgoing local interface 1



BGP, OSPF, forwarding table entries

Q: how does router set forwarding table entry to distant prefix?

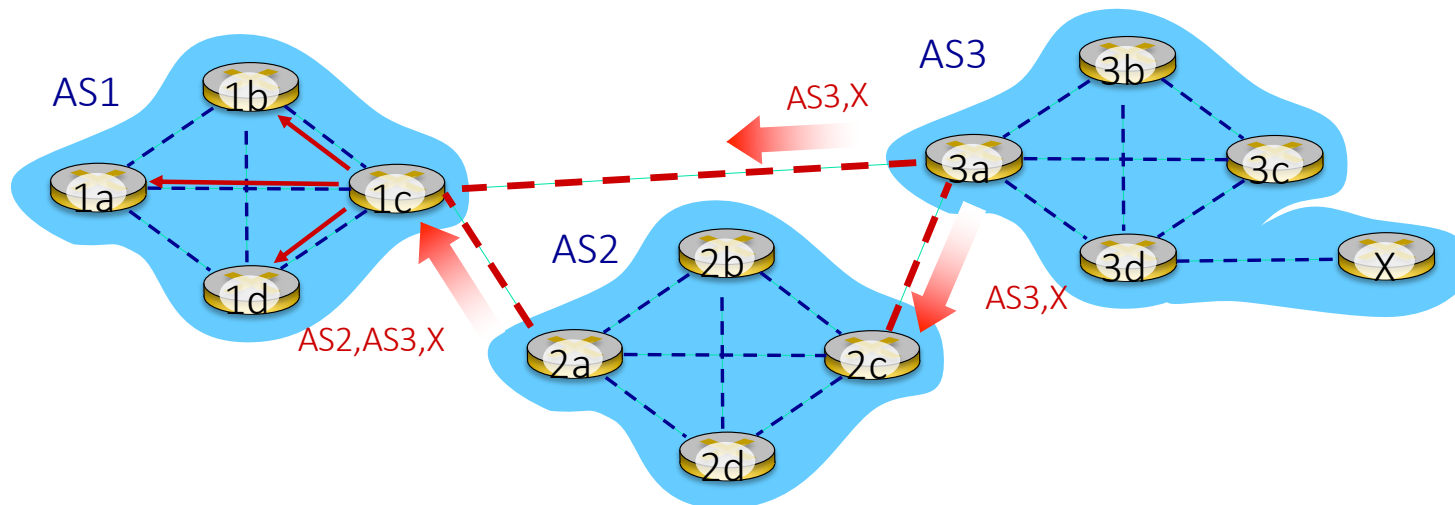


- recall: 1a, 1b, 1c learn about dest X via iBGP from 1c: “path to X goes through 1c”
- 1d: OSPF intra-domain routing: to get to 1c, forward over outgoing local interface 1
- 1a: OSPF intra-domain routing: to get to 1c, forward over outgoing local interface 2



BGP path advertisement

- gateway router may learn about multiple paths to destination:
 - AS1 gateway router 1c learns path AS2,AS3,X from 2a
 - AS1 gateway router 1c learns path AS3,X from 3a
 - Based on policy, AS1 gateway router 1c chooses path AS3,X, and advertises path within AS1 via iBGP



BGP route selection

- router may learn about more than one route to destination AS, selects route based on:
 1. local preference value attribute: policy decision
 2. shortest AS-PATH
 3. closest NEXT-HOP router: hot potato routing
 4. additional criteria

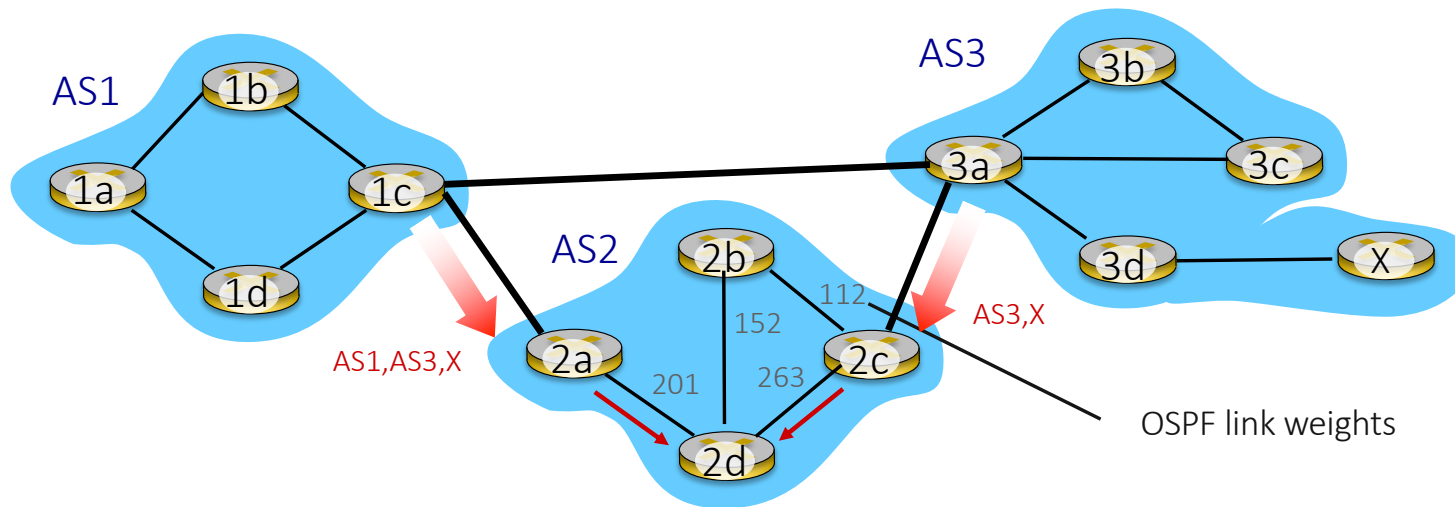


Problems: BGP Hijacking

- A **BGP hijack** happens when an AS falsely originates someone else's IP prefix
 - other networks believe that AS is the “right” path for that traffic.
- The attacker (or misconfigured AS) announces a prefix, often **more specific than the legitimate one** (e.g., /24 vs /23), and BGP's “longest prefix match” makes most of the Internet send traffic to the wrong place.
- Common causes
 - Misconfiguration (wrong prefix, missing filters), compromised routing/registry accounts, or deliberate malicious attacks to intercept, inspect, or blackhole traffic.
 - Famous 2008 Pakistani incident involving YouTube
- Typical effects
 - Traffic detoured through the hijacker, leading to outages, performance degradation, eavesdropping/Man-in-the-Middle, or full blackholing of a service or region.
- Mitigations: Prefix-filtering by upstreams, origin validation, max-prefix / max-length checks, and better operational hygiene at ISPs and IXPs.



Hot Potato Routing



- 2d learns (via iBGP) it can route to X via 2a or 2c
- *hot potato routing*: choose local gateway that has least intra-domain cost (e.g., 2d chooses 2a, even though more AS hops to X): don't worry about inter-domain cost!
- BGP would however would have selected (AS3, X) based on criterium 2
 - Suboptimal routing



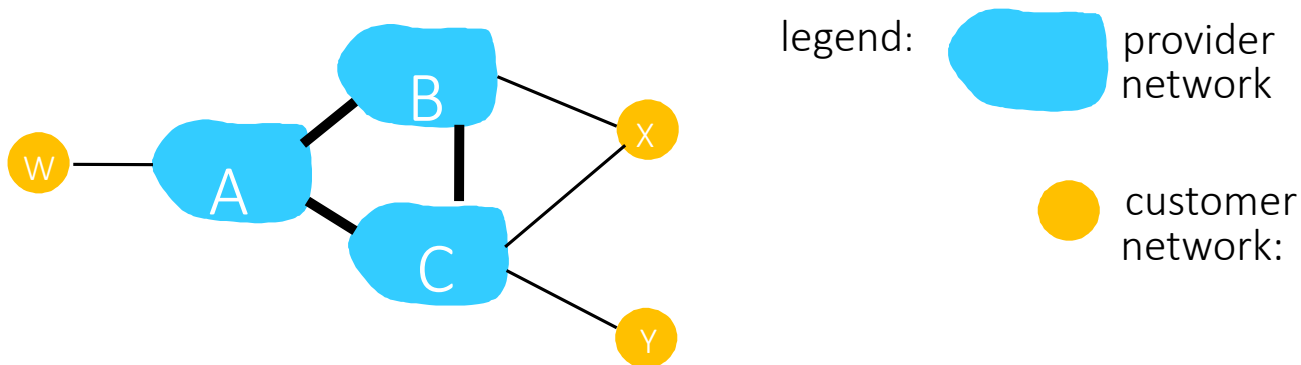
BGP Connection Types

- There are two types of interconnections:
- **Transit**: an ISP can provide reachability to the entire Internet for another endpoint (e.g., enterprise, content or application provider, residential broadband provider, etc.).
 - The endpoint entity pays the ISP to carry traffic to and from the Internet.
- **Peering**: The networks interconnect to exchange only traffic that originates or terminates within their networks (including the networks of their customers, in the case of carriers and Tier 1 networks)
 - **Public peering** through an internet exchange point (IXP) -> Peering with *multiple* networks
 - **Private peering**: networks interconnect to exchange only traffic that originates or terminates within their networks



BGP: achieving policy via advertisements

- Let us look at a case of *Private Peering*
 - A,B,C are provider networks
 - X,W,Y are customer (of provider networks)
 - X is dual-homed: attached to two provider networks
 - Policy to enforce: **X does not want to route from B to C via X** -> Private peering
 - .. so X will not advertise to B a route to C



BGP: achieving policy via advertisements

- Let us now look at a case of *Peering Agreements* among ISPs
 - A advertises path A_w to B and to C
 - B chooses not to advertise B_{A_w} to C:
 - B gets no “revenue” for routing C_{B_{A_w}}, since none of C, A, w are B’s customers
 - C does not learn about C_{B_{A_w}} path
 - C will route C_{A_w} (not using B) to get to w
- Rule of thumb: traffic flowing across an ISP’s backbone network must have either a source or a destination (or both) in a network that is a customer of that ISP;

